

CE 310 / Correlation

When Variables Move Together

Week 7 · Class 1 · The Building as a System of Relationships

University of Arizona · Dr. Wooyoung Jung · Fall 2026

Part 1

A Confounding Mistake

The Daylighting Trap

What happened — Phoenix, 2017:

A building owner installed a full-glass south facade to reduce lighting energy. Bills showed lighting was highest in summer and lowest in winter.

Conclusion: "Daylighting reduces lighting in winter but is ineffective in summer."

Reality: No daylight sensors. The seasonal pattern came from **occupancy hours** — longer winter days. Both variables were driven by the same confounder. Cost: \$85,000 in upgrades with zero effect.

The lesson:

Correlation describes a statistical pattern in two measured variables. It does not reveal which variable causes which, or whether a third hidden variable — a **confounder** — is driving both. Always ask: is a third variable driving both?

Week 6 Looking Ahead — Answer

Moving together, or just sharing a spreadsheet?

Week 6 asked: **How do you tell whether two variables actually move together, versus just happening to sit in the same spreadsheet?**

Answer: The **Pearson correlation coefficient** r — one number from `CORREL`, ranging -1 to $+1$.

Same file, same 8,760 hours, same building. Outdoor temperature vs. cooling energy: the Pearson $r = 0.7752$ — hotter hours, higher cooling loads.

Zone temperature vs. cooling energy: $r = 0.2764$. Sharing a spreadsheet is not a relationship.

Today's Tools — Five Concepts

By the end of this lecture you will be able to:

1. **Pearson r** — measure strength and direction of linear co-movement (`CORREL` function)
2. r^2 — fraction of one variable's variance explained by the other
3. **Covariance** — the raw joint variation before standardizing (`COVARIANCE.S` function)
4. **Conditional correlation** — r computed on a subgroup (season, operating mode)
5. **Scatter plot** — visual tool for seeing the shape of the relationship

This week's dataset: `Week5_D0E_Med0ffice_ZoneTemp_2023.csv` — download from D2L (8,760 hourly rows)

- Column G = `Zone_Temp_F` · Column H = `Outdoor_Temp_F` · Column I = `Cooling_kWh`

Part 2

Measuring Co-Movement

What Correlation Captures

What r measures:

- Direction of the linear relationship (positive or negative)
- How consistently two variables move together
- Dimensionless — no units
- Ranges from -1 (perfect negative) to $+1$ (perfect positive)
- $r = 0$ means no linear relationship

What r does NOT tell you:

- Which variable causes which
- Whether the relationship is non-linear (a curved relationship can have $r \approx 0$)
- Whether a third variable is driving both
- What happens on any individual hour

A strong r does not mean causation. It means two variables tend to move together in the data.

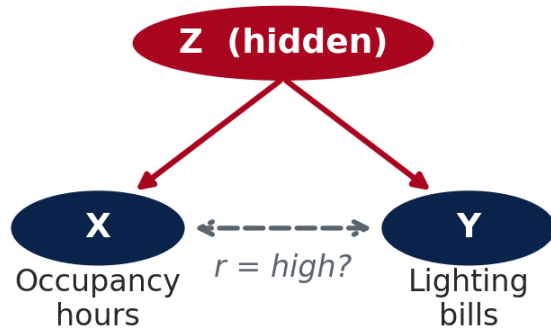
Correlation $\neq$ Causation — Three Structures

1. Confounder — a hidden Z drives both X and Y . Remove Z and the apparent X - Y correlation weakens or disappears. (*Daylighting trap: occupancy hours drove both lighting bills and apparent daylighting performance.*)

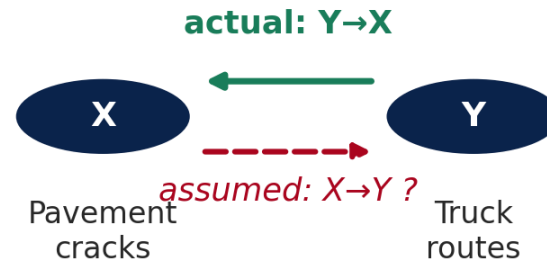
2. Reverse causation — $Y \rightarrow X$, not $X \rightarrow Y$. The correlation is real, but the assumed direction is backward. (*Heavy trucks correlate with pavement damage — but trucks cause the cracks, not the reverse.*)

Three Structural Reasons $r \neq$ Causation

1. Confounder
 Z drives both — no $X \rightarrow Y$ link

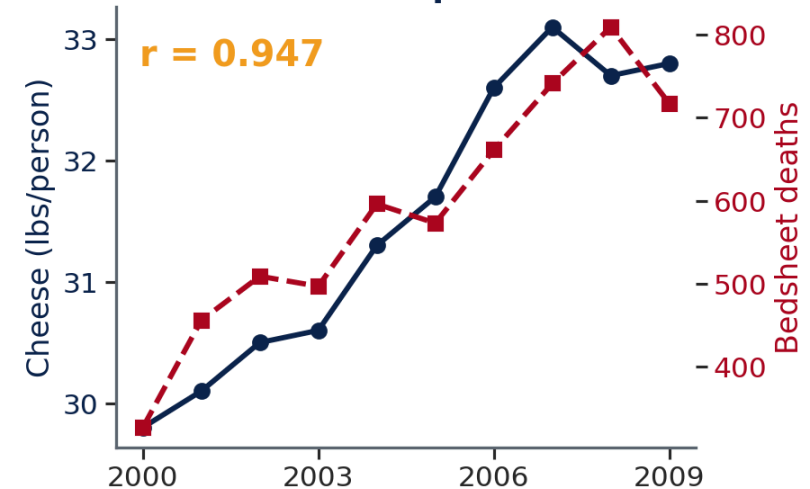


2. Reverse Causation
 $Y \rightarrow X$, not $X \rightarrow Y$



"Cracks attract trucks?" No — trucks create cracks.

3. Coincidence
Two unrelated upward trends



Correlation $\neq$ Causation — Three Structures (cont.)

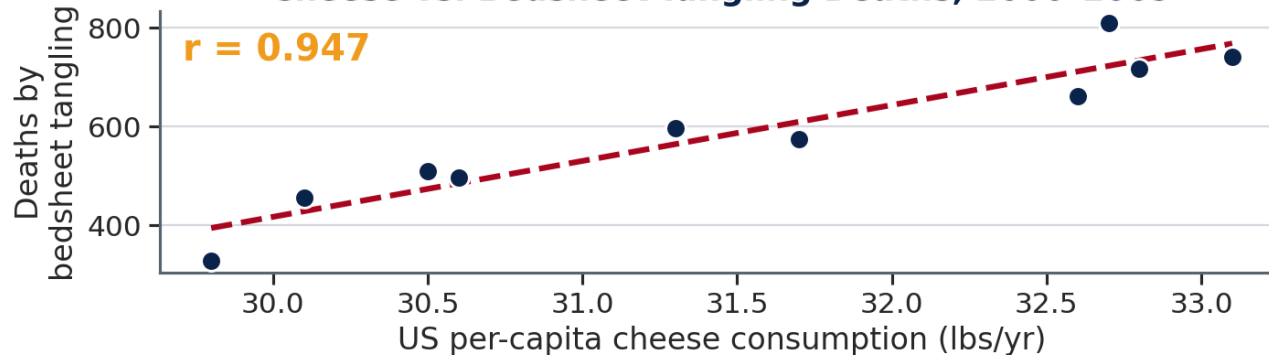
3. Coincidence — two unrelated variables both trend over the same period. $r = 0.95$, mechanism = none.

High r cannot tell you which of these three is true. Only domain knowledge — the physical mechanism — resolves it.

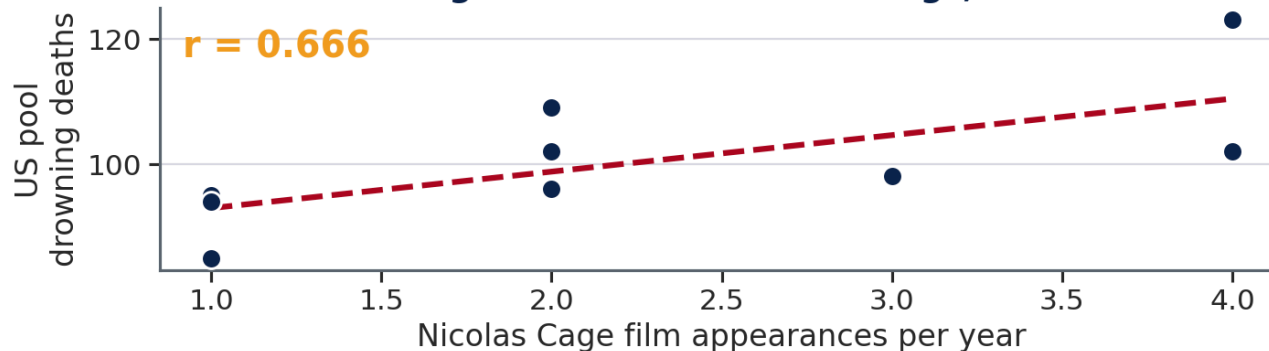
Famous Spurious Correlations

High r — Zero Mechanism (Tyler Vigen, 2015; USDA; CDC)

Cheese vs. Bedsheet-Tangling Deaths, 2000-2009



Nicolas Cage Films vs. Pool Drownings, 1999-2009



Both pairs have **strong Pearson r** — the cheese–bedsheet pair ($r = 0.947$) exceeds our own outdoor-cooling pair ($r = 0.775$).

Neither has a physical mechanism.

Why? Any two series that both trend upward over the same years produce a high r — the algorithm cannot tell a physical link from a coincidence of trends.

💡 **The question r cannot answer:** "Is there a mechanism by which X physically produces Y ?" — engineering judgment, not Excel.

The Pearson r Formula

$$r = \frac{\text{cov}(X, Y)}{\sigma_X \cdot \sigma_Y}$$

Numerator: covariance — how X and Y vary together (has units)

Denominator: product of standard deviations — removes units, scales result to $[-1, +1]$

Covariance — what the numerator measures:

$$\text{cov}(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

When X and Y are both above their means simultaneously $\rightarrow$ positive product $\rightarrow$ positive contribution. Summed over all n observations, the sign and size capture the dominant direction of co-movement. This is the variance formula, generalized to two variables.

The Pearson *r* Formula (cont.)

Applied to outdoor temperature and cooling energy:

$$r = \frac{331.58 \text{ }^{\circ}\text{F}\cdot\text{kWh}}{16.918 \text{ }^{\circ}\text{F} \times 25.282 \text{ kWh}} = \frac{331.58}{427.6} = 0.7752$$

Function	Result
=CORREL(X, Y)	0.7752 — use this
=PEARSON(X, Y)	0.7752 — identical; CORREL is the course convention

How to Read r

Abs(r)	Strength	Engineering meaning
0.9 – 1.0	Very strong	Nearly perfect linear predictor
0.7 – 0.9	Strong	Reliable for preliminary models
0.5 – 0.7	Moderate	Useful but significant unexplained variation
0.3 – 0.5	Weak	Poor linear predictor
0.0 – 0.3	Negligible	No linear relationship

Sign: Positive $r \rightarrow Y$ rises with X ; Negative $r \rightarrow Y$ falls with X .

$r = 0.7752$: **strong positive** — outdoor temperature is a reliable predictor of cooling load.

Part 3

Three Pairs, One Surprise

Correlation Across CE Domains

Pearson r is a universal tool — the same formula applies across every CE domain:

Domain	Variable Pair	Typical r	Direction	Physical reason
Materials / Structural	Water-cement ratio → 28-day compressive strength	≈ -0.90	Negative	More water dilutes cement paste → weaker concrete (ACI 318)
Transportation	Freeway vehicle density → average speed	≈ -0.75	Negative	More vehicles → congestion → slower flow (HCM Level of Service)
Hydrology	Rainfall intensity → runoff coefficient	$\approx +0.82$	Positive	Faster rain exceeds soil infiltration rate → more runoff (NRCS TR-55)
Building Energy	Outdoor temp → cooling load	0.78	Positive	Hotter weather → more HVAC work (this week's dataset)

Correlation Across CE Domains (cont.)

The sign and magnitude of r encode the physical mechanism. A plausible mechanism is also what separates a useful engineering model from a spurious coincidence.

Three Correlation Pairs

Pair	r	Interpretation
Outdoor_Temp_F vs. Cooling_kWh	0.7752	Strong positive — main pair for regression
Zone_Temp_F vs. Outdoor_Temp_F	0.7529	Moderate-strong — HVAC limits zone temperature swing
Zone_Temp_F vs. Cooling_kWh	0.2764	Weak — zone temp is a poor predictor of cooling load

Largest r : Outdoor vs. Cooling (0.7752). **Weakest:** Zone vs. Cooling (0.28) — not the pair most students expect.

Students often expect zone-cooling to be the strongest pair: "if the zone is hot, HVAC must be working hard." The data says otherwise. The next slide explains why.

Why Zone-Cooling r is Weak — HVAC Control

Intuition: "Hot zone $\rightarrow$ high cooling?" Direction yes ($r = +0.28$), but r is weak because zone temp barely varies ($\sigma = 2.92^\circ\text{F}$). The thermostat clamps zone temp — zone temp is the **result** of HVAC winning.

$r(\text{Outdoor, Cooling}) = 0.7752$

Outdoor temp swings 16.9°F — from 28°F January nights to 110°F July afternoons. Large variance $\rightarrow$ strong signal.

$r(\text{Zone, Cooling}) = 0.2764$

Zone temp constrained to $\pm 2.9^\circ\text{F}$ around setpoint. Little variance $\rightarrow$ little signal.

Zone temp is a **thermostat success metric** — tight control makes it a poor predictor of HVAC effort.

r^2 — The Coefficient of Determination

r^2 is the fraction of variance in Y explained by X :

$$r^2 = \frac{[\text{cov}(X, Y)]^2}{\text{var}(X) \cdot \text{var}(Y)}$$

Pair	r	r^2	Interpretation
Outdoor vs. Cooling	0.7752	0.6010	Outdoor temp explains 60.1% of cooling variance
Zone vs. Cooling	0.2764	0.0764	Zone temp explains only 7.6% of cooling variance

Outdoor temp is roughly **8x more informative** than zone temp (60.1% vs. 7.6%).

The remaining 39.9% unexplained by outdoor temp alone includes:

- Solar gain variation (cloud cover, orientation)

Quick Check — Correlation and Causation

Think, then check with your neighbor

Question A: If $r(\text{Outdoor, Zone}) = 0.75$ and $r(\text{Outdoor, Cooling}) = 0.78$, does that mean outdoor temperature is a *cause* of both zone temperature and cooling energy?

Question B: A building in Las Vegas has $r(\text{Outdoor, Cooling}) = 0.92$; for a building in Seattle, $r = 0.61$. Which climate has a more consistent outdoor-cooling relationship, and why?

Compare with your neighbor.

✓ Correlation and Causation — Answers

Question A: Correlation does NOT imply causation — but outdoor temperature DOES physically drive cooling through thermodynamics. The r value tells you only about statistical co-movement. The physical mechanism (conduction, convection, solar gain) provides the causal link, but r alone cannot establish it. Both statements can be true simultaneously: high r AND a real causal mechanism.

Question B: Las Vegas has a more consistent outdoor-cooling relationship because it operates in a single dominant regime — hot, dry, high solar gain year-round. Seattle's moderate climate has mixed heating/cooling seasons, maritime cloud variability, and mild shoulder seasons that add scatter to the outdoor-cooling relationship.

Climate	$r(\text{Outdoor, Cooling})$	Why
Las Vegas (hypothetical)	0.92	Single dominant cooling regime
Tucson (actual)	0.78	Mixed: cooling season + ~3,700 chiller-off hours

Part 4

When r Changes by Subgroup

Why Condition?

The all-hours $r(\text{Outdoor, Cooling}) = 0.7752$ mixes 8,760 hours with very different operating modes: occupied vs. unoccupied, summer vs. winter, business hours vs. setback.

Subgroup	$r(\text{Outdoor, Cooling})$	n	What this means
All hours	0.7752	8,760	Blended average of all modes
Business hours	0.9539	2,600	Nearly perfect — active control mode
Non-business hours	0.6864	6,160	Weaker — setback reduces correlation
Spring	0.8143	2,208	Cleanest season (no monsoon scatter)
Summer	0.7494	2,208	Lowest seasonal r (monsoon adds scatter)

Business Hours — r Jumps to 0.95

All hours: $r = 0.7752$, $n = 8,760$

Includes 6,160 non-business hours where the HVAC is in setback mode — cooling turns off or reduces regardless of outdoor temperature, **breaking the linear relationship.**

During setback, the outdoor-cooling link is weak: the system is not responding dynamically to outdoor conditions.

Business hours only ($F = \text{"Yes"}$): $r = 0.9539$, $n = 2,600$

During occupied hours, HVAC responds dynamically to outdoor temperature within the thermostat band. Hot outdoor day → high cooling demand → strong linear signal.

$r^2 = 0.9100$: outdoor temperature explains **91% of cooling variance** during business hours.

The all-hours $r = 0.78$ answers "what is r for the full year?" — not "what is r during active HVAC mode?" Choose based on your design question.

Seasonal r — Spring Highest, Summer Lowest

Season	r	n	Why
Spring	0.8143	2,208	Clean cool-to-warm sweep — near-monotone, strong linear signal
Fall	0.7524	2,184	Symmetric to spring — similarly clean sweep
Winter	0.7541	2,160	Real cooling still occurs from internal gains
Summer	0.7494	2,208	LOWEST — monsoon humidity and solar gain add scatter

Range: $\text{MAX} - \text{MIN} = 0.8143 - 0.7494 = 0.0649$

Only 0.065 range across all four seasons — the outdoor-cooling r is essentially the same year-round. One regression model (Week 9) works for all seasons without seasonal adjustment.

Simpson's Paradox – The Negative Summer r

$r(\text{Zone_Temp_F}, \text{Outdoor_Temp_F})$ for **summer only** = -0.2452

All hours: $r(\text{Zone}, \text{Outdoor}) = +0.7529$ – positive; zone and outdoor generally move together across seasons.

Summer only: $r(\text{Zone}, \text{Outdoor}) = -0.2452$ – **negative**; within summer, zone temp falls as outdoor temp rises.

This is Simpson's Paradox: the direction of a relationship can **reverse** when you condition on a subgroup.

Why the inversion occurs:

In Tucson summer at 108°F, the HVAC runs at maximum capacity, actively **driving zone temperature down**. On the hottest days, the HVAC fights hardest – inverting the expected relationship.

Season	$r(\text{Zone}, \text{Outdoor})$	Regime
Summer	-0.2452	HVAC dominates

Month is Not a Physical Predictor

$$r(\text{Month}, \text{Outdoor_Temp_F}) = 0.1438 \cdot r(\text{Month}, \text{Cooling_kWh}) = 0.0942$$

Both near zero — yet we know temperature is strongly seasonal. **Why?**

Pearson r only detects **monotone linear** relationships. Outdoor temperature in Tucson rises January → July then falls July → December. This parabolic (up-then-down) shape averages out to near-zero linear correlation with month (1–12).

Predictor	r with Cooling_kWh	r^2	Use in regression?
Month (1–12)	0.0942	0.009	No — non-linear pattern invisible to Pearson r
Outdoor_Temp_F	0.7752	0.601	Yes — physical mechanism, strong linear signal

Always use the **physical mechanism variable**, not a calendar label. Month encodes seasonality indirectly and non-linearly; outdoor temperature encodes the same information directly and linearly.

Quick Check B — Simpson's Paradox and Non-Linearity

Think, then check with your neighbor

Question A: $r(\text{Zone}, \text{Outdoor})$ for all hours is $+0.75$, but for summer it is -0.25 . Can both be right at the same time?

Question B: If a colleague says "the correlation between month and cooling is only 0.09, so there's no seasonality," how would you respond?

Compare with your neighbor.



Quick Check B — Simpson's Paradox

Both r values are correct — they answer different questions

Question A: Yes — both can be correct at once. The all-hours $r = +0.75$ is driven by the seasonal mean difference: winter zone temps cluster at 68°F, summer at 75°F, so the two trend together across the year. Within summer, the HVAC control reversal dominates — when outdoor temp spikes, the chiller runs harder and zone temp stays flat or drops below setpoint.

Question B: Month as a predictor fails because its relationship with cooling is **parabolic (U-shaped rotated 90°)**: cooling peaks in July, bottoms in December/January. Pearson r only detects monotone linear trends — it cannot find a pattern that goes up then comes back down. The correct tools are AVERAGEIF by month, or a scatter plot of monthly average cooling vs. month number.



When $r \approx 0$, ask: "Could the relationship be non-linear or non-monotone?"

Part 5

From Covariance to Correlation

Covariance — The Raw Signal

$$\text{cov}(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

- When X and Y are **both above** their means: positive product → positive contribution
- When X above and Y below their means: negative product → negative contribution
- Sum of all products → positive cov means they tend to move together

Applied to outdoor temperature vs. cooling:

$$\text{cov}(\text{Outdoor}, \text{Cooling}) = 331.58 \text{ } ^\circ\text{F}\cdot\text{kWh}$$

Units: $^\circ\text{F}\cdot\text{kWh}$ — the product of both variables' units. Correlation divides out the units:

$$r = \frac{331.58}{16.918 \times 25.282} = \frac{331.58}{427.6} = 0.7752\checkmark$$

Part 6

AI / ML Bridge

Correlation in Machine Learning

Three ML algorithms where correlation is the core mathematical operation:

Linear Regression

(Week 9)

$$\beta_1 = r \cdot \frac{\sigma_Y}{\sigma_X}$$

The regression slope **is** correlation, rescaled to the variables' units.

CE: outdoor temp → cooling load model

Principal Component Analysis (PCA)

When predictors are correlated (temperature, humidity, solar gain...), PCA rotates the **correlation matrix** into independent axes — the first captures maximum shared variance.

CE: energy benchmarking — reduce 30 correlated metrics to 3 independent axes

Collaborative Filtering

"Customers who bought X also bought Y." User similarity = Pearson r of rating vectors. Users whose ratings co-move ($r \approx 1$) likely agree on items neither has rated yet.

CE analogy: recommending standard spec sections based on what similar project types used

Multicollinearity and Feature Selection

In machine learning and regression modeling, computing r between all input features is the first step in **exploratory data analysis (EDA)**.

Rule: if two features have $|r| > 0.9$, they carry nearly identical information — including both creates **multicollinearity**: inflated standard errors, unstable coefficient estimates, unreliable predictions.

Feature pair	Abs(r)	Multicollinearity risk
Outdoor_Temp_F vs. Month	0.14	None — essentially independent as predictors
Zone_Temp_F vs. Outdoor_Temp_F	0.75	Moderate — some overlap but tolerable
Outdoor_Temp_F vs. Cooling_kWh	0.78	Not applicable — predictor vs. response

Week 9 uses outdoor temperature alone — not because Month creates multicollinearity, but because it adds almost no predictive power (0.9% vs. outdoor's 60.1%).

Looking Ahead — Week 8: Midterm Review

Week 8 is the **midterm**. No new dataset, no new Excel function.

Conceptual questions:

- Interpreting r , r^2 , compression ratio
- HVAC control mode effects on correlation
- Simpson's Paradox, confounder identification

All Weeks 1–7 Excel functions are fair game.

See Tutorial Quick Reference slides for the complete list.

Formula writing (apply to dataset):

- NORM.DIST patterns (CDF vs. PDF, upper-tail)
- At-least- k Poisson formulas
- CORREL with subgroup FILTER
- COVARIANCE.S syntax
- Conditional probability from COUNTIFS

Key Results — All-Hours Correlations

Quantity	Value
$r(\text{Zone_Temp_F}, \text{Outdoor_Temp_F})$ — all hours	0.7529
$r(\text{Outdoor_Temp_F}, \text{Cooling_kWh})$ — all hours	0.7752
$r(\text{Zone_Temp_F}, \text{Cooling_kWh})$ — all hours	0.2764
$r^2(\text{Outdoor_Temp_F}, \text{Cooling_kWh})$	0.6010

Key Results — Conditional Correlations

Quantity	Value
$r(\text{Outdoor_Temp_F}, \text{Cooling_kWh})$ — business hours	0.9539
$r(\text{Outdoor_Temp_F}, \text{Cooling_kWh})$ — non-business hours	0.6864
$r(\text{Zone_Temp_F}, \text{Outdoor_Temp_F})$ — Summer only	−0.2452
$r(\text{Zone_Temp_F}, \text{Outdoor_Temp_F})$ — Winter only	0.7837
$\text{cov}(\text{Outdoor_Temp_F}, \text{Cooling_kWh})$	331.58 °F·kWh
$\text{cov}(\text{Zone_Temp_F}, \text{Outdoor_Temp_F})$	37.15 °F ²

Week 7 Lab — Correlation

Week7_Lab_Instructions.pdf · Week7_Answer_Sheet.xlsx

Section A — Pearson r and r^2 :

- A1–A3: CORREL for all three variable pairs (outdoor-cooling, zone-outdoor, zone-cooling)
- A4: Multiple choice — which pair has the largest $|r|$
- A5: r^2 for the main pair (outdoor vs. cooling)
- A6: Conditional CORREL for business hours using FILTER
- A7–A8: Multiple choice — interpreting r and r^2 values

Section B — Covariance and Conditional Correlation:

- B1–B5: Seasonal r for each season using FILTER